The documentation and process conversion measures necessary to comply with this revision shall be completed by 3 February 2007.

INCH-POUND

MIL-PRF-19500/359G
3 November 2006
SUPERSEDING
MIL-PRF-19500/359F
22 March 2002

* PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, FAST RECOVERY, POWER RECTIFIER, 1N4942, 1N4944, 1N4946, 1N4947, AND 1N4948, JAN, JANTX, AND JANTXV

Inactive for new design after 27 February 1992 For new design use - 1N5615, 1N5617, 1N5619, 1N5621, 1N5623 on MIL-PRF-19500/429.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

* The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a silicon, fast recovery semiconductor power rectifier diode for use in equipment circuits. Three levels of product assurance are provided for each device as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to DO-41).

* 1.3 Maximum ratings.

Device type	V_{RWM} V_R	I_O (1) (2) T_A = +55°C	I_O (2) T_A = +100°C	I_{FSM} at t_p = 8.3 ms T_A = +100°C	Barometric pressure	t_{rr1} at I_F = 0.5 A, I_R = 1.0 A I_{REC} = 0.25 A	T_{STG} and T_J
	V pk (V_{RWM}) V dc (V_R)	A dc	A dc	A pk	mm Hg	ns	°C
1N4942	200	1.0	.750	15		150	-65 to +175
1N4944	400	1.0	.750	15	8	150	-65 to +175
1N4946	600	1.0	.750	15	8	250	-65 to +175
1N4947	800	1.0	.750	15	33	250	-65 to +175
1N4948	1,000	1.0	.750	15	33	500	-65 to +175

(1) Derate linearly from 1.0 A at T_A = +55°C to 0.75 A at +100°C. Derate linearly from 0.75 A to 0 A between +100°C and +175°C.
(2) No forced air nor heat sinking shall be permitted for I_O ratings at T_A = +55°C and +100°C.

* Comments, suggestions, or questions on this document should be addressed to Defense Supply Center, Columbus, ATTN: DSCC-VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dscc.dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

* 2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil. or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

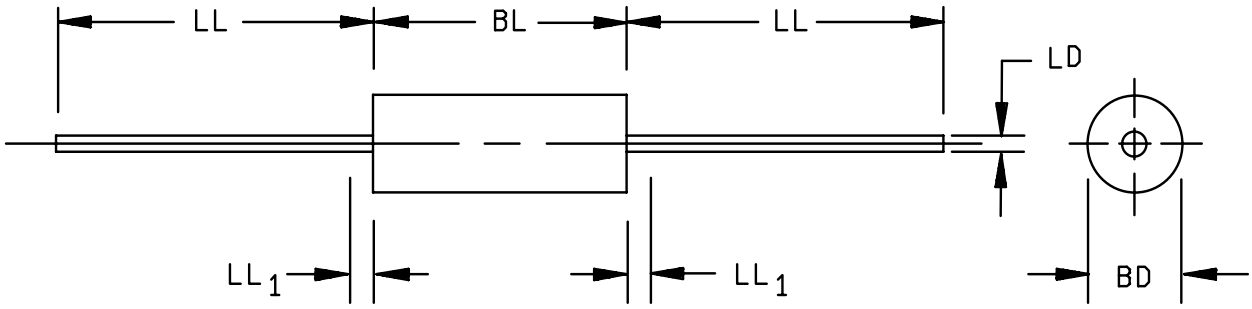
* 3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

* 3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

T_{CVF} ..... Temperature coefficient of forward voltage.

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Dimensions					
Ltr	Inches		Millimeters		Notes
	Min	Max	Min	Max	
BD	.065	.150	1.65	3.81	3, 4
BL	.140	.250	3.56	6.35	3
LD	.027	.033	0.69	0.84	
LL	1.00	1.50	25.4	38.1	
LL1		.050		1.27	5

NOTES:
1. Dimensions are in inches.
2. Millimeter equivalents are given for general information only.
3. Dimension BL and BD include all components of the diode periphery except the section of leads over which the diameter is controlled.
* 4. Dimension BD shall include the sections of the lead over which the diameter is uncontrolled. This uncontrolled area is defined as the zone between the edge of the diode body and extending .050 inch (1.27 mm) onto the leads.
5. Lead diameter not controlled in this zone to allow for flash, lead finish build-up, and minor irregularities other than slugs.
6. In accordance with ASME Y14.5M, diameters are equivalent to ϕX symbology.

* FIGURE 1. Physical dimensions (similar to DO-41).

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3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 herein. All devices shall be metallurgically bonded-thermally matched-noncavity-double plug construction as defined in MIL-PRF-19500.

3.4.1 Encapsulant material. In addition to those categories of hermetically sealed package requirements specified in MIL-PRF-19500, fused metal oxide-to-metal shall also be acceptable.

3.4.2 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

* 3.5.1 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end.

* 3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

* 3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

* 3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

* 4.1.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-19500 and herein.

* 4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and herein.

* 4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table E-IV of MIL-PRF-19500)	Measurement
	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2) - army
9	Not applicable
11	I_{R1} and V_{FM1}
12	See 4.3.1 and 4.5.3
13	Subgroup 2 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or $\pm$50 nA dc whichever is greater. $\Delta V_{FM1} \leq$ +0.1 V dc, -0.2 V dc. Scope display evaluation (see 4.5.4).

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows (see 4.5.3.1): $I_{O(min)}$ = 1 A dc. T_A = +55°C maximum. Test conditions in accordance with method 1038 of MIL-STD-750, condition B. Adjust I_O or T_A to achieve the required T_J. T_J = 135°C minimum. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

* 4.3.1 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 or 4081 of MIL-STD 750 as applicable, using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (V_C and V_H where appropriate). Measurement delay time (t_{MD}) = 70 µs max. The thermal impedance limit used in screen 3c and table I, subgroup 2 shall be set statistically by the supplier.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500 and table I herein.

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* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta requirement shall be in accordance with table III herein.

* 4.4.2.1 Group B inspection, appendix E, table E-VIb (JAN, JANTX, and JANTXV of MIL-PRF-19500).

Subgroup	Method	Condition
B2	4066	I_{FSM} = rated (see 1.3); ten surges of 8.3 ms each at 1 minute intervals; I_O = 0.75 A dc; V_{RSM} = 0.
B3	1027	I_O = 1.0 A dc minimum; f = 50 - 60 Hz; T_A = +50°C max.; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.3.1); t = 340 hours.
B5	3101 or 4081	$R_{\theta JL}$ = 38°C/W; L = .375 inch (9.52 millimeters); I_M = 10 mA, t_{md} = 100 µs, I_h = 2 A, Th = thermal equilibrium.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta requirements shall be in accordance with table III herein.

* 4.4.3.1 Group C inspection, appendix E, table E-VII of MIL-PRF-19500.

Subgroup	Method	Condition
C2	1056	Test condition A.
C2	2036	Tension: Test condition A, weight = 5 pounds; t = 15 seconds.
C2	1021	Omit initial conditioning.
C5		Not applicable.
C6	1026	T_A = +50°C max. f = 50 - 60 Hz; I_O = 1.0 A minimum; V_R = rated V_{RWM} (see 1.3, 4.5.3, and 4.5.3.1)

* 4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IX of MIL-PRF-19500 and as specified in table II herein. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta requirements shall be in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

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4.5.3 Burn-in and life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees nor less than 150 degrees.

4.5.3.1 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_O is increased such that the junction temperature of each diode is maintained at +135°C minimum for screening and +150°C for life test.

4.5.4 Scope display evaluation. Scope display evaluation shall be sharp and stable in accordance with method 4023 of MIL-STD-750. Scope display may be performed on ATE (automatic test equipment) for screening only with the approval of the qualifying activity. Scope display in group A shall be performed on a scope.

4.5.5 Peak reverse power test. This test shall be measured in the circuit as shown on figure 2, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2						
Forward voltage	4011	I_{FM1} = 1.0 A dc, pulsed (see 4.5.1).	V_{FM1}	0.6	1.3	V dc
Reverse current leakage	4016	V_R = rated V_{RWM}, (see 1.3), pulsed (see 4.5.1).	I_{R1}		1.0	µA
Breakdown voltage	4021	I_R = 50 µA, pulsed (see 4.5.1).	$V_{(BR)1}$			
1N4942				220		V dc
1N4944				440		V dc
1N4946				660		V dc
1N4947				880		V dc
1N4948				1100		V dc
Thermal impedance	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Subgroup 3						
High temperature operation		T_A = +150°C				
Reverse current leakage	4016	V_R = rated V_{RWM} (see 1.3), pulsed (see 4.5.1).	I_{R2}		200	µA
Low temperature operation		T_A = -65°C				
Forward voltage	4011	I_{FM} = 1 A dc, pulsed (see 4.5.1).	V_{FM2}	0.6	1.5	V dc
Reverse current	4016	V_R = rated V_{RWM} (see 1.3); dc method.	I_{R3}		1.0	µA dc

See footnote at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Forward recovery voltage	4026	I_F = 0.25 A; t_p = 20 ns, (minimum); t_r = 8 ns.	V_{fr}		5.0	V
Capacitance	4001	V_R = 12 V dc, f = 0.1 to 1 MHz.	C_J			
1N4942					45	pF
1N4944					35	pF
1N4946					25	pF
1N4947					20	pF
1N4948					15	pF
Reverse recovery time	4031	Condition B1.	t_{rr1}			
1N4942					150	ns
1N4944					150	ns
1N4946					250	ns
1N4947					250	ns
1N4948					500	ns
Scope display	4023	Stable, see 4.5.4 n = 116, c = 0				
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-PRF-19500.

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* TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices c = 0
Temperature cycling	1051	500 cycles, condition A.	
Hermetic seal	1071		
Fine leak		Test condition H. 5×10^{-7} atm cc/s	
Gross leak		Test condition C or K	
Electrical measurement		See table I, group A, subgroup 2.	
Subgroup 2			22 devices c = 0
Steady-state reverse bias	1038	Test condition A t = 1,000 hours, T_C = +125°C V_R = 0.8 of rated V_{RWM} (see 1.3).	
Subgroup 3			
Not applicable			
Subgroup 4			Sample size N/A
Thermal impedance		See MIL-PRF-19500	
Subgroup 5			22 devices c = 0
Barometric pressure (reduced)	1001	1N4944, 1N4946 = 8 mm Hg (100,000 ft); 1N4947, 1N4948 = 33 mm Hg (70,000 ft); Voltage during test = V_{RWM}; maximum; Leakage (I_R) during test shall be 5 μA.	
Subgroup 6			
Not applicable			
Subgroup 7			10 devices c = 0
Peak reverse power		See 4.5.5 and figure 2 herein. $P_{RM} \geq 500$ W. Test shall be performed on each sublot	
Electrical measurement		See table I, group A, subgroup 2.	

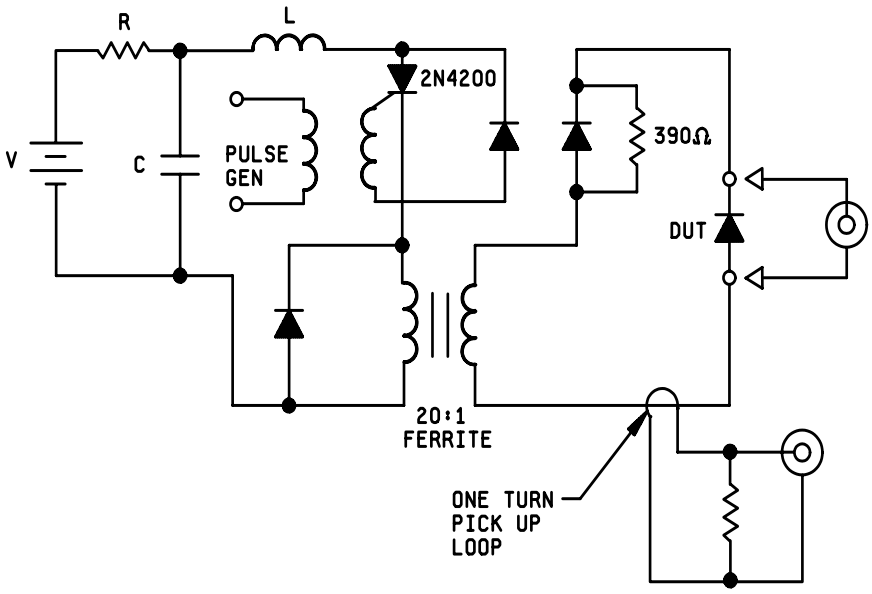
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* TABLE III. Groups B, C, and E delta measurements. 1/ 2/ 3/ 4/

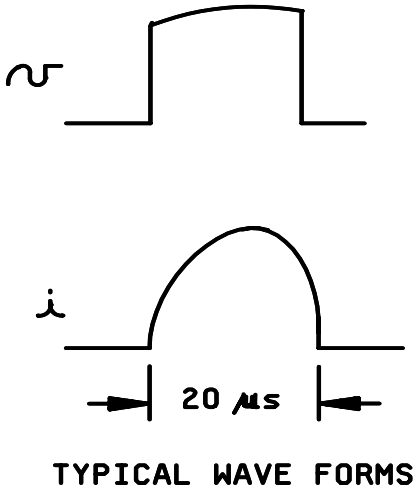
Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	I_{FM1} = 1.0 A dc, pulsed (see 4.5.1).	ΔV_F	-0.2	+0.1	V dc
2.	Reverse current leakage	4016	V_R = rated V_{RWM} , (see 1.3), pulsed (see 4.5.1).	ΔI_R	100 percent of initial value or 50 nA, whichever is greater		

1/ Devices which exceed the table I limits for this test shall not be accepted.
2/ The electrical measurements for appendix E, table E-VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table III herein, step 2.
3/ The electrical measurements for appendix E, table E-VII of MIL-PRF-19500 are as follows:
   a. Subgroup 2, see table III herein, steps 1 and 2.
   b. Subgroup 6, see table III, steps 1 and 2.
4/ The electrical measurements for appendix E, table E-IX of MIL-PRF-19500 are as follows:
   a. Subgroup 1, see table III herein, steps 1 and 2.
   b. Subgroup 2, see table III herein, steps 1 and 2.

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NOTES:
1. L - 13T #22 pm 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 μf to give a 20 μs pulse width.
3. V - adjustable to 200 volts for power desired in D.U.T.



TYPICAL WAVE FORMS

FIGURE 2. Peak reverse power measurement circuit and waveform.

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5. PACKAGING

* 5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

* 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The notes specified in MIL-PRF-19500 are applicable to this specification.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.2).

d. Product assurance level and type designator.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil.

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN (Part or Identifying Number). This information in no way implies that manufacturers' PIN's are suitable as substitutes for the military PIN's.

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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